

How the data-quality assessment of the OPERA archive was computed

Notes for the `minkowski-loss` project

23 September 2026

Abstract

These notes describe the data-quality audit run on 23 September 2026 over the raw OPERA radar archive: 745 days, 4.8 million tiles of 128×128 pixels. They cover every step from the stored rain rate to the summary tables:

- what is read;
- how tiles are cut and keyed;
- how a per-pixel clutter climatology is built;
- the exact definition of each of the ~ 45 per-tile features;
- the candidate artefact rules;
- the statistics in the summary;
- the split-leakage audit;
- the image gallery used for labelling.

Each quantity is defined mathematically and illustrated. A final part reports the first full-run results and how the code was checked.

The audit *describes* the data. It deletes nothing, and its rule thresholds are starting guesses to be calibrated against labels.

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1 What the assessment is for

The project’s claims are about the upper tail of rain rate. Artefacts of the radar network concentrate in that tail: ground and sea clutter, interference spokes, range rings, one-frame bursts and decoding errors. A patch ranked by its maximum is therefore more likely to be an artefact the higher it ranks. The audit answers four questions before any screening decision is made:

1. Which artefact *signatures* are present, and how common are they as a function of intensity, year, season, time of day and location?
2. Which ones overlap, so that a single rule would suffice, and which ones are independent?
3. What does the current preprocessing (`filter_precip_bounds`) do to the tail?
4. How independent are the current train/validation/test splits?

Everything is computed on the *raw* stored field, before any filtering, and on the full downloaded archive, not only on the patches that entered training.

2 The data it reads

Grid and time. Each day is one zarr store `RAW_OPERA_DATA_DIR/YYYYMMDD/`. It holds a rain-rate array $R_t(y, x)$ in mm/h, for t over the day’s 15-minute composites and (y, x) on the OPERA 2 km Lambert azimuthal equal-area grid of 2200×1900 pixels. Pixels outside radar coverage are NaN, and “no echo” is 0. Stores written by the archive fetcher (`scripts/data/fetch_opera_archive.py`) also carry the OPERA total quality index $Q_t(y, x) \in [0, 1]$. Figure 1 shows the coverage.

Two kinds of store. The raw directory mixes two sources, and they differ in a way that matters.

- **Original 2023–24 stores** (Aug 2023 – Oct 2024, 442 days): the source of the current patch set. They hold the rate only, no QIND.
- **Fetched archive stores** (2012–13 and test days in 2016, 2020 and July 2023, 303 days): written by the fetcher. They hold rate and QIND, with no alteration of values.

Figure 2 compares their tile maxima. The original stores *never exceed 150 mm/h*. Over their 2,853,816 fully covered tiles the maximum is exactly 150.0 and the 99.99th percentile is 149.94. The archive stores continue smoothly past 150 and reach $\sim 10^5$ mm/h (decoding errors and clutter).

- So the 150 mm/h cut is present in the original raw source itself, not only in our preprocessing. Whether the source clipped or removed the values above 150 cannot be told from the stores.
- Recovering real 2023–24 peaks would need those days re-fetched from the archive.

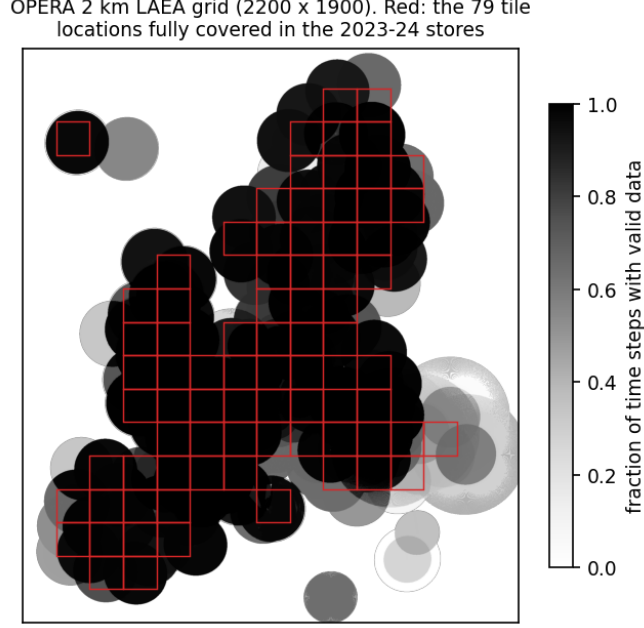


Figure 1: Fraction of time steps with valid data at each pixel over all 745 days (grey), and the tile locations that are fully covered in the 2023–24 stores (red). Tiles are 128×128 pixels (256×256 km) on a stride-128 grid.

3 Pipeline overview

Stage by stage (all scripts are in `scripts/data_quality/`; the feature definitions are in `src/data/quality.py`):

1. `clutter_climatology.py` counts, per pixel, how often each threshold is exceeded (§5).
2. `compute_tile_features.py` cuts every tile at every time step and computes the features (§6). Clutter frequencies from stage 1 are joined at the tile’s peak.
3. `audit_splits.py` measures the overlap of the current splits (§9).
4. `summarize_features.py` applies the candidate rules and aggregates (§7–8).
5. `patch_gallery.py` renders tiles for inspection and writes label sheets (§10).

4 Tiles and keys

A tile is identified by its time stamp t and its lower-left pixel (r, c) , with r, c multiples of 128. It is the array

$$T(i, j) = R_t(r + i, c + j), \quad 0 \leq i, j < 128. \quad (1)$$

The tile grid, the key format `timestamp,row,col` and the orientation are exactly those of `scripts/preprocess/generate_metadata.py`. So every feature row joins one-to-one onto the existing patch metadata. A tile is kept when its coverage $\text{cov}(T) = \frac{1}{128^2} \sum_{ij} \mathbf{1}[T(i, j) \text{ finite}]$ equals 1 (flag `--min_coverage`), which is the current patch definition.

Remark (Alignment check). Applying the pipeline’s own filter to a tile and taking the maximum reproduces the `max_precip` column of the metadata exactly (difference 0.0 on every joined test tile). The number of fully covered tiles in the original stores, 2,853,816, equals the total number of train + validation + test patches. Both checks confirm that the audit sees exactly the tiles the model was trained on.

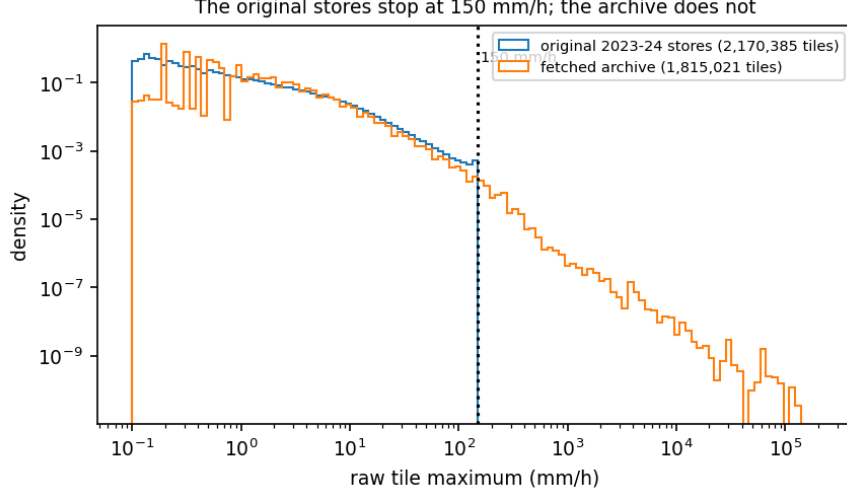


Figure 2: Distribution of raw tile maxima in the two kinds of store (tiles with $\max > 0.1$ mm/h).

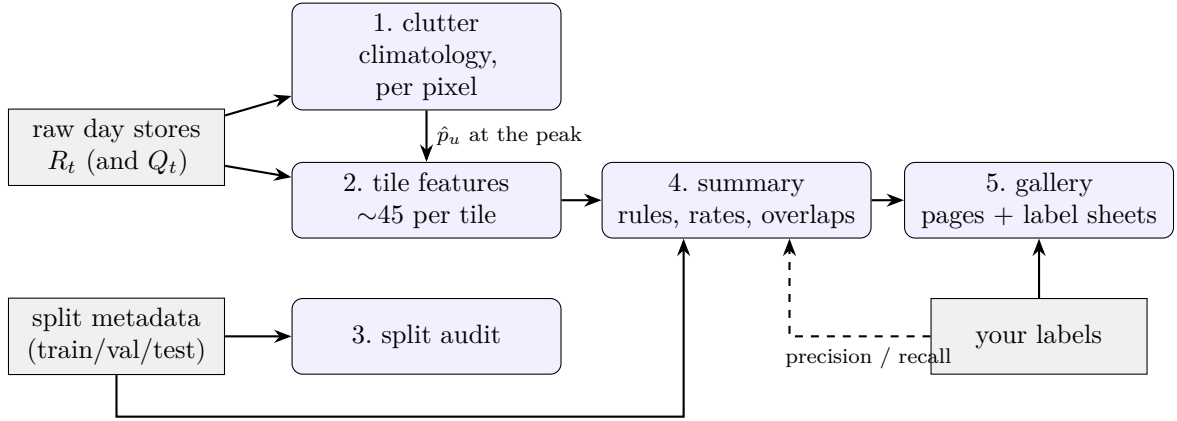


Figure 3: The five stages of `scripts/hpc/launch_data_quality.sh`. Dashed: the labelling loop, which closes after you label tiles.

5 Stage 1: the clutter climatology

A real storm crosses a given pixel a handful of times a year. Ground clutter, wind farms, sea clutter and interference light up the *same* pixels again and again. For every pixel and threshold $u \in \{1, 31, 89, 150, 500\}$ mm/h, over all time steps of all days:

$$n(y, x) = \sum_t \mathbf{1}[R_t(y, x) \text{ finite}], \quad (2)$$

$$n_u(y, x) = \sum_t \mathbf{1}[R_t(y, x) \geq u], \quad \hat{p}_u(y, x) = \frac{n_u(y, x)}{\max(n(y, x), 1)}. \quad (3)$$

The estimate $\hat{p}_u$ is the empirical exceedance frequency. Weather varies smoothly in space on these scales, while clutter is pixel-sharp. The *local excess*

$$e_u(y, x) = \hat{p}_u(y, x) - \text{med}_{15 \times 15} \hat{p}_u(y, x) \quad (4)$$

therefore removes the climatological background (a 15×15 median filter, 30 km) and keeps what stands out from its surroundings. The script also stores yearly counts, the per-pixel maximum and the mean QIND, and it ranks the top-50 pixels by e_{31} (`clutter_hotspots.csv`). Figure 4

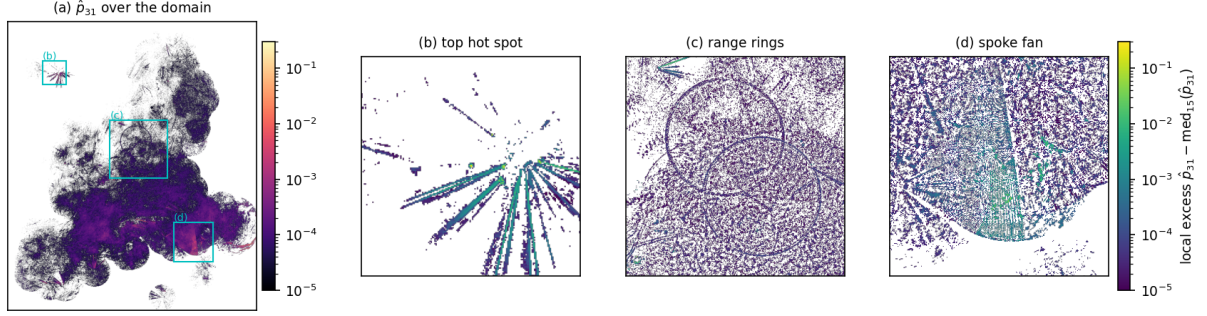


Figure 4: (a) Exceedance frequency of 31 mm/h over all 745 days. (b–d) Local excess e_{31} in three zooms. The top hot-spot pixel exceeds 31 mm/h in 23% of all time steps.

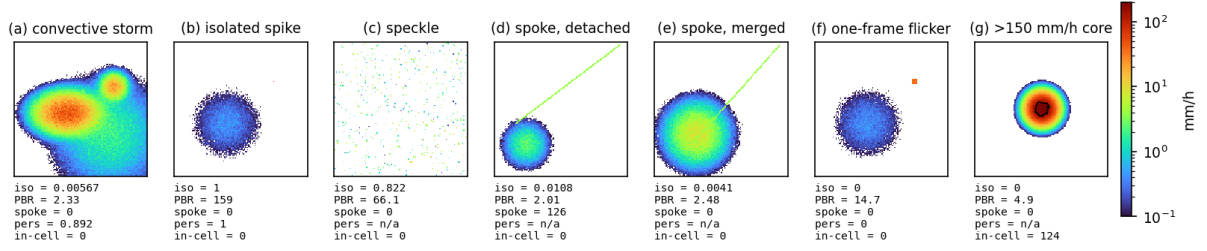


Figure 5: Synthetic tiles and the values the audit computes for them. **iso**: isolated fraction at 1 mm/h. **PBR**: peak/coarse-block ratio. **spoke**: spoke length in px. **pers**: persistence into $t - 15$. **in-cell**: pixels > 150 mm/h inside a cell. Panel (e) shows a known limitation: a spoke that touches rain merges with it and is not detected.

shows the result: sharp spokes radiating from individual radars, complete range rings, and a fan of interference.

The tile features join two numbers from this map (§6.9): $\hat{p}_u$ at the tile’s peak pixel, and the maximum of $\hat{p}_u$ over the tile.

6 Stage 2: the tile features

All features are computed on the raw tile T , with NaN treated as missing. The notation used throughout:

- $Z = T$ with NaN $\rightarrow 0$;
- $M = \max Z$, the tile maximum, attained at $p^* = (i^*, j^*)$;
- $\mathcal{N}_8(p)$, the eight neighbours of pixel p (edge pixels have fewer);
- $\delta = 0.1$ mm/h, the drizzle level;
- $D = 150$ mm/h, the current declutter threshold.

Figure 5 applies the actual feature code to synthetic tiles that isolate one signature each.

6.1 Intensity and concentration

$$M = \max_{ij} Z_{ij}, \quad \bar{T} = \text{mean}_{\text{finite}} T, \quad q_{99} = 99\text{th percentile}, \quad (5)$$

$$\text{wet} = \frac{1}{|T|} \#\{T \geq \delta\}, \quad N_u = \#\{T \geq u\} \quad (u \in \{1, 31, 53, 89, 150, 500\}), \quad (6)$$

$$\text{conc} = \bar{T}/M. \quad (7)$$

The *concentration* is near 0 when the tile’s rain is carried by a few pixels. A real convective

cell raises M and the wet area together, so conc falls only slowly with M . A single bright pixel drives it to $\sim 10^{-4}$.

6.2 Speckle, spikes and gradients

Let $\text{nbmax}(p) = \max_{q \in \mathcal{N}_8(p)} Z_q$, with nbmean , nbmed and nbmin defined likewise.

- **Isolated fraction** at level $u \in \{0.1, 1, 10\}$: the share of pixels $\geq u$ with no neighbour $\geq u$,

$$\text{iso}_u = \frac{\#\{p : Z_p \geq u, \text{nbmax}(p) < u\}}{\#\{p : Z_p \geq u\}}. \quad (8)$$

Rain fields are spatially coherent at 2 km, so iso_u is small for weather and large for speckle (Fig. 5c: 0.82).

- **Spikes**: pixels ≥ 10 mm/h whose brightest neighbour is below a quarter of them, $\mathcal{S} = \{p : Z_p \geq 10, \text{nbmax}(p) < \frac{1}{4}Z_p\}$. The features are $n_{\text{spikes}} = |\mathcal{S}|$ and the *spike mass fraction* $\sum_{\mathcal{S}} Z / \sum Z$.
- **Peak-to-neighbour ratio**: $M / (\text{nbmean}(p^*) + \delta)$.
- **Maximum gradient**: the largest absolute difference between 4-neighbours, $\max |T_{i+1,j} - T_{i,j}| \vee |T_{i,j+1} - T_{i,j}|$, in mm/h per 2 km.

6.3 Connected components and the spoke detector

Components are the 8-connected sets of $\{Z \geq u\}$ for $u \in \{1, 31\}$. The features are the number of components and the size of the largest.

Interference spokes are long, thin and straight. For each component C with $|C| \geq 30$ px, take the covariance of its pixel coordinates,

$$\Sigma_C = \frac{1}{|C|} \sum_{p \in C} (p - \bar{p})(p - \bar{p})^\top, \quad \lambda_{\max} \geq \lambda_{\min} \text{ its eigenvalues.} \quad (9)$$

The *elongation* $\epsilon_C = 1 - \sqrt{\lambda_{\min}/\lambda_{\max}}$ is 0 for a disc and tends to 1 for a line. A uniform segment of length L has variance $L^2/12$ along its axis, so

$$L_C = \sqrt{12 \lambda_{\max}} \quad (10)$$

estimates the component's length. The features are $\max_C \epsilon_C$ and the *spoke length* $\max\{L_C : \epsilon_C > 0.9\}$ (Fig. 5d: 126 px).

Remark. A spoke crossing a rain area is part of a larger, non-elongated component and is invisible to this detector (Fig. 5e). Detecting it would need radar-site geometry (azimuth from the site), which the audit does not have.

6.4 Relation to the coarse input

The model is conditioned on a 10×10 coarse field made by `torch.nn.functional.adaptive_avg_pool2d` of the *filtered* tile. With $n = 128$ and $m = 10$, bin k covers pixels

$$[\lfloor kn/m \rfloor, \lceil (k+1)n/m \rceil), \quad (11)$$

which are 13 or 14 pixels wide and overlap by one (Fig. 6). The audit reproduces these bins exactly (unit-tested against torch). Let $\bar{B}(p^*)$ be the mean of the coarse block containing the peak. Then

$$\text{PBR} = \frac{M}{\bar{B}(p^*) + \delta}. \quad (12)$$

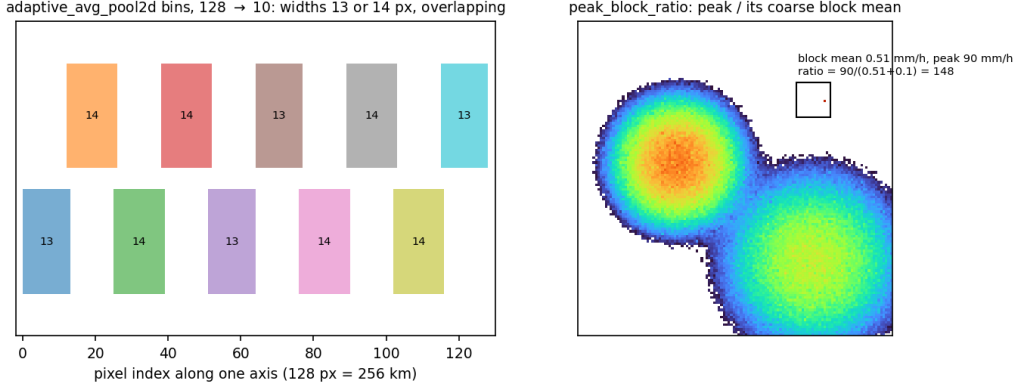


Figure 6: Left: the ten adaptive-pooling bins along one axis. Right: the peak/coarse-block ratio for an isolated 90 mm/h pixel.

A lone pixel in an otherwise dry block gives $\text{PBR} \approx$ the block area (~ 170). A convective core, whose surroundings are also wet, gives single digits. A large PBR means the target contains a peak that the input gives no hint of, which is exactly what makes artefacts harmful to a super-resolution model.

6.5 What the current declutter step does

`filter_precip_bounds` sets every value outside $[\delta, D]$ to zero, and NaN to zero. The audit counts

$$n_{>D} = \#\{p : Z_p > D\}, \quad n_{>D}^{\text{cell}} = \#\{p : Z_p > D, \text{nbmed}(p) \geq 10\}, \quad (13)$$

the second being the pixels above 150 that sit *inside* a wet area. Zeroing those punches a hole in the storm (Fig. 7, top). It also counts pre-existing holes, $\#\{p : Z_p < \delta, \text{nbmin}(p) \geq 10\}$.

6.6 Terrain and sea

With the DEM on the same grid, sea is $\{\text{DEM} \leq 0\} \cup \{\text{DEM missing}\}$. The features are:

- the sea fraction and mean elevation;
- whether the peak lies over sea;
- the share of wet (≥ 1 mm/h) pixels that lie over sea.

Sea clutter shows up as intense peaks over sea in coarse blocks that are almost dry.

6.7 OPERA's quality index

Where the store has QIND:

- its coverage $\#\{Q \text{ finite}\} / \#\{T \text{ finite}\}$;
- the mean of Q over all pixels, over wet pixels and over pixels ≥ 31 mm/h;
- the share of ≥ 31 mm/h pixels with $Q < 0.5$;
- $Q(p^*)$.

The QIND coverage is itself informative: in the archive it is missing on most of the covered area.

6.8 Persistence in time

Real storms persist and move. At 15-minute resolution a cell moving at ≤ 50 km/h travels ≤ 12.5 km, about 6 pixels. With W the 13×13 window centred on p^* (± 6 px), and $R_{t\pm 1}$ the neighbouring frames of the same tile:

$$\text{pers}_{\pm} = \frac{\max_W R_{t\pm 1}}{M}. \quad (14)$$

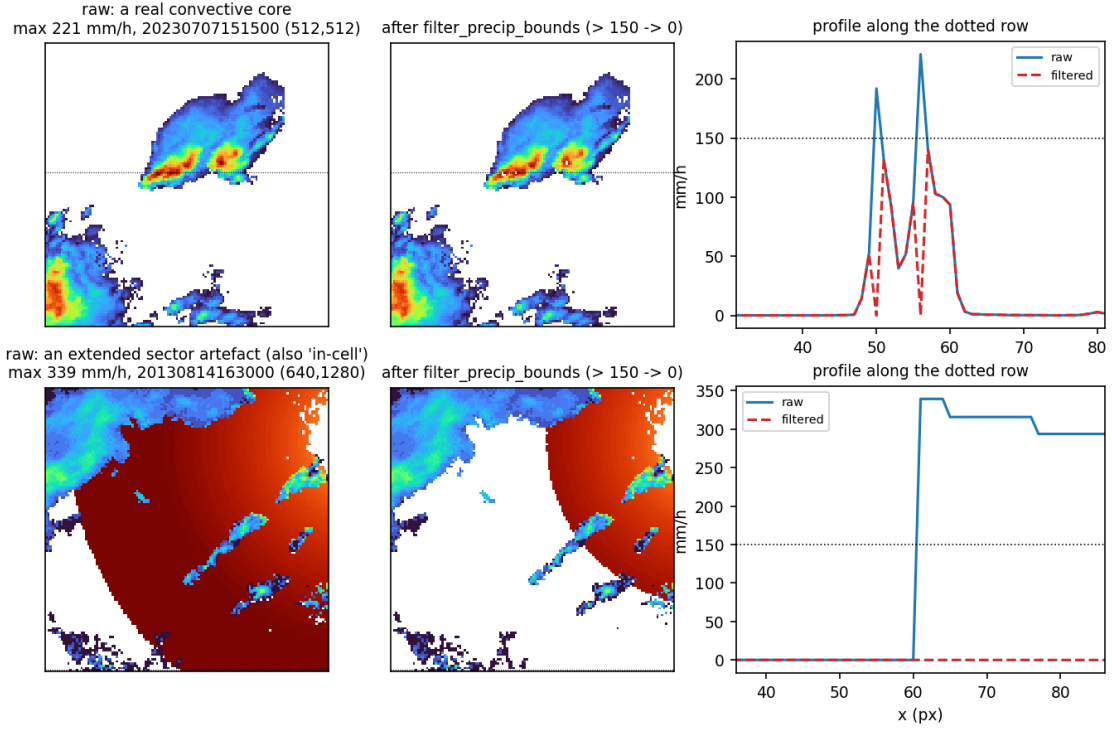


Figure 7: Top: a real convective core from the fetched July 2023 data. Its two peaks above 150 mm/h are set to 0 by the current filter, leaving holes at the storm’s centre. Bottom: an extended high-value sector artefact. It also passes the “inside a cell” test, because its neighbours are high too. The in-cell count therefore identifies holes in real storms only together with the other features.

A persistent storm gives values near 1: the median for tail tiles with a large core in 2024 is 0.88. A one-frame burst gives ≈ 0 in both directions (Fig. 5f).

Values well above 1 are also informative: the neighbouring frame holds something much brighter nearby, typically a burst in *that* frame. The Pearson correlation of $\log(1 + R_t)$ with $\log(1 + R_{t-1})$ over the tile is recorded as `corr_prev`. The first and last frames of a day have no neighbour across the day boundary, and their persistence is missing.

6.9 Static clutter at the peak

From stage 1: $\hat{p}_u(p^*)$ at the tile’s peak pixel, and $\max_T \hat{p}_u$ over the tile, for each u . A tile peaking on a pixel that exceeds 31 mm/h in more than 1% of all time steps is almost certainly peaking on clutter.

A caveat: this is only as good as the climatology. On a few days it flags recurring *weather* too. Over 745 days it isolates the static artefacts.

7 Stage 4a: candidate rules

Each rule turns features into a flag for one signature. **The thresholds are first guesses**, to be calibrated against labels (§10).

rule	flag condition
spike	$n_{\text{spikes}} > 0$ and spike mass fraction > 0.2
speckle	$\text{iso}_1 > 0.5$ and $N_1 < 200$
isolated_peak	$M \geq 10$ and $\text{PBR} > 60$
flicker	$M \geq 10$ and $\max(\text{pers}_-, \text{pers}_+) < 0.2$
sea_clutter	$M \geq 10$, peak over sea, and $\bar{B}(p^*) < 0.5$
spoke	spoke length > 40 px (80 km)
static_clutter	$M \geq 31$ and $\hat{p}_{31}(p^*) > 0.01$
qind_low	$M \geq 10$ and $Q(p^*) < 0.3$
declutter_isolated	$n_{>D} > n_{>D}^{\text{cell}}$ (pixels > 150 outside any wet area)
unphysical	$M > 500$ mm/h
declutter_hole	$n_{>D}^{\text{cell}} > 0$: not an artefact flag; the tiles the current filter would hole

Missing inputs are not “no”. A rule whose inputs are missing gives NaN, not 0. Examples are `qind_low` on stores without QIND, and `flicker` at the first frame of a day. Rates are then computed over tiles where the rule is defined. `any_flag` is 1 if any defined rule fires, and missing only if no rule is defined.

8 Stage 4b: the summary statistics

Strata. Tiles are grouped by raw maximum M into $[0, 0.1)$, $[0.1, 1)$, $[1, 10)$, $[10, 31)$, $[31, 53)$, $[53, 89)$, $[89, 150)$, $[150, 500)$ and $[500, \infty)$ mm/h. The *tail* is $M \geq 31$ mm/h, the POT level of the project’s tail protocol. Stratifying by the raw maximum defines the tail before anything is removed.

Flag rates. For a group G and rule k , $\hat{\pi}_k(G) = \frac{1}{|G_k|} \sum_{G_k} f_k$ over the tiles $G_k \subseteq G$ where f_k is defined. The binomial standard error $\sqrt{\hat{\pi}(1 - \hat{\pi})/|G_k|}$ understates the true uncertainty: tiles of the same storm and the same hour are strongly correlated. Treat differences below a few percentage points as noise until they are recomputed with day-level resampling.

The key diagnostic. In a clean archive, artefact signatures would become *rarer* towards the tail, because real extremes are large coherent storms. A flag rate that rises with M is therefore evidence of contamination, whatever the rules’ individual precision.

Overlap. Rules k, l are compared by the Jaccard index over the tail, $J_{kl} = |F_k \cap F_l|/|F_k \cup F_l|$, where F_k is the set of flagged tiles. Small off-diagonal values mean the rules catch different things, so a screen needs their union.

Breakdowns and outputs. Rates are also broken down by year, month, hour (UTC), current split and tile location. The summary lists every flagged tail tile (`flagged_tail.csv`, 216,714 tiles) for the gallery.

Against labels. Given a sheet of labelled tiles (artefact / real / unsure), each rule gets

$$\text{precision} = \frac{TP}{TP + FP}, \quad \text{recall} = \frac{TP}{TP + FN} \quad (15)$$

over tiles labelled artefact or real. Labels must come from a *random* draw of the tail as well as from flagged tiles. Otherwise precision cannot be separated from the base rate.

9 Stage 3: the split audit

`split_metadata.py` shuffles individual patches, so the same storm can be in train and test. For each test (and validation) patch (t, r, c) the audit asks whether the train split contains:

- *same time*: any patch at the same t ;
- *same tile within 1 h*: a patch at (t', r, c) with $|t' - t| \leq 60$ min (found by binary search in the tile’s sorted train times);
- *adjacent tile, same time*: a patch at $(t, r \pm 128, c \pm 128)$.

query	subset	n	same time	same tile ± 1 h	adjacent, same t
test	all	285,383	100.0%	100.0%	98.2%
test	patch max ≥ 31	25,591	100.0%	100.0%	99.4%
test	patch max ≥ 89	6,349	100.0%	100.0%	99.3%

Validation gives the same numbers. Every current test score is therefore an interpolation score within storms seen in training.

10 Stage 5: the gallery and the labelling loop

`patch_gallery.py` renders one row per tile (Fig. 8):

- the raw frames at $t - 15$, t and $t + 15$ min;
- the target after the current filter, where holes show as white;
- the coarse input as the model sees it;
- the DEM and QIND.

The row header carries the key, the raw maximum and the rules that fire. The selections are:

- **top_max**: the highest maxima;
- **rule:<name>**: the tiles each rule flags;
- **unflagged**: tail tiles no rule flags, to find misses;
- **random**: an unbiased draw from the tail, needed for precision and recall.

Each page set comes with `labels.csv`. Fill its `label` column and pass the file to `summarize_features.py --labels` to close the loop in Fig. 3.

11 First full run: what it shows

The full run took about 4 hours with 6 workers on 8 cores. It covered 745 days (2012-09-04 to 2024-10-30) and 4,804,051 tiles, of which 413,654 (8.61%) are in the tail.

1. **The flag rate rises with intensity** (Fig. 9a). Any-flag is 18% of tiles at 1–10 mm/h, 47% at 31–53, 54% at 89–150, and $\approx 100\%$ above 150. That is the contamination signature of §8.
2. **Old years are dirtier** (Fig. 9b). 92% of 2012 tail tiles are flagged, 75% in 2013, and 42% and 37% in 2023 and 2024. The QIND-based rule exists only for the fetched years.
3. **A diurnal cycle** (Fig. 9c). The tail flag rate is $\approx 61\%$ at night and $\approx 43\%$ mid-afternoon. Two effects combine: nighttime clutter (anomalous propagation under stable layers) and afternoon convection adding real tail tiles.
4. **The rules are complementary**. Most tail Jaccard indices are below 0.3; the largest is `isolated_peak` with `flicker` (0.36). No single rule is a screen.
5. **Static clutter is real and located** (Fig. 4). A few tile locations have 85–97% of their tail tiles flagged. The worst is (1024, 640).

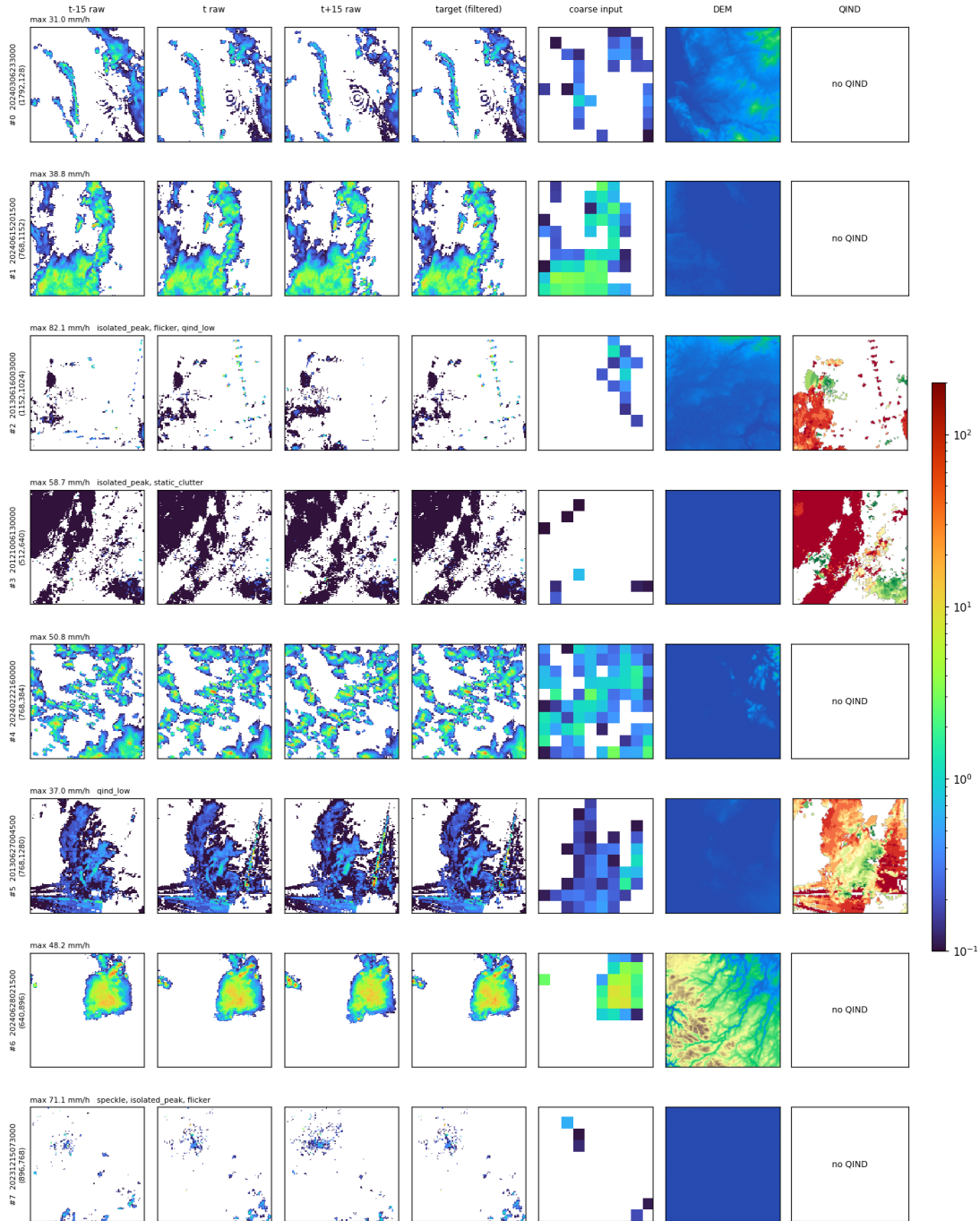


Figure 8: First page of the random tail draw. Rows 1, 4 and 6 are real storms with no rule firing. Rows 2, 3 and 7 are flagged artefacts. Row 0 is a real storm with a ring artefact that no rule catches. Row 5 is a storm crossed by a spoke that merges with the rain. Only `qind_low` fires there; the spoke rule misses it (Fig. 5e).

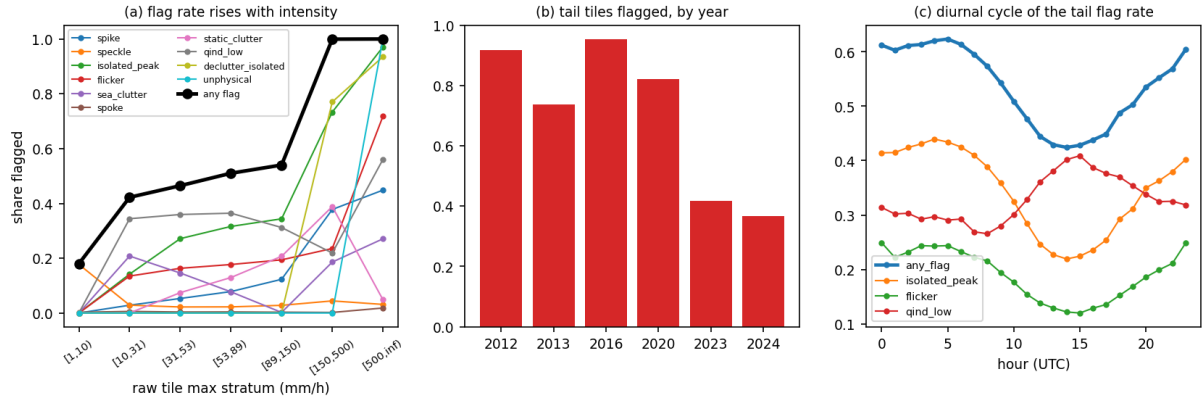


Figure 9: (a) Flag rate by raw-max stratum, per rule and for any rule. (b) Share of tail tiles flagged, by year. (c) Diurnal cycle of the tail flag rate.

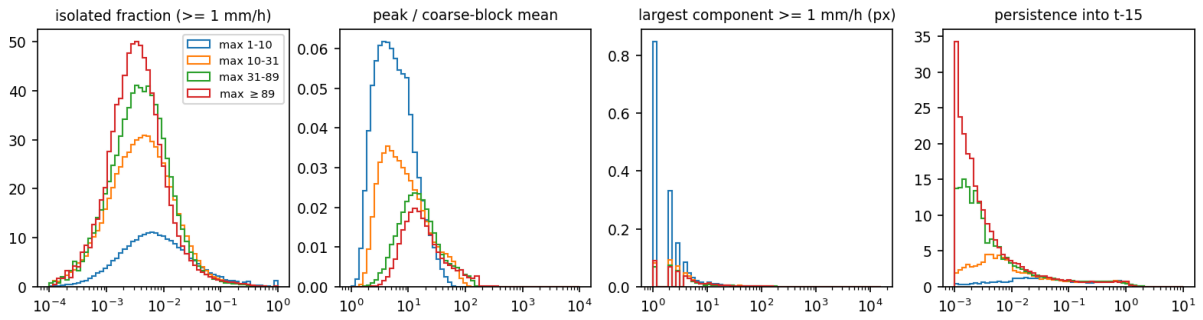


Figure 10: Four features by raw-max stratum (tiles with $\max \geq 1$ mm/h; log-spaced bins, density per bin). Persistence has a long low tail (the flicker cases, $\approx 10\%$ of the tail). Log-spaced bins plotted as density visually exaggerate that tail. The median persistence is 0.53–0.72 in every stratum.

6. **The declutter step touches 5.7% of tail tiles.** It would hole real cells in 7,867 of them, all in the fetched years, since the original stores stop at 150.

12 How the code was checked

- **Unit tests** (tests/test_quality.py, 8 tests). Each artefact feature must separate its synthetic artefact from a synthetic storm: spike against storm, the spoke's length within 5%, flicker against advection, and declutter-in-cell against isolated. The coarse pooling is checked against `adaptive_avg_pool2d` to 10^{-6} , and NaN and all-NaN tiles are handled.
- **Alignment.** The filtered tile maximum reproduces the metadata `max_precip` exactly. The original-store tile count equals the patch count.
- **Smoke tests** on 5 days before the full run, with visual inspection of the gallery pages.
- **Sanity checks on the full run.**
 - The distributions behave as expected by stratum (Fig. 10).
 - The persistence of large real cores has a median of 0.88.
 - Extreme persistence ratios ($> 10^3$, 65 tiles) trace to bursts in the neighbouring frame, not to arithmetic errors.

13 Limitations

- Rule thresholds are uncalibrated. The rates measure signature prevalence, not artefact counts.
- There is no radar-site geometry, so spokes and rings merged with rain are missed (Fig. 5e, Fig. 8, rows 0 and 5).
- The in-cell criterion for > 150 mm/h pixels is fooled by extended artefacts (Fig. 7, bottom).
- Flag rates are correlated across tiles of the same storm, so the standard errors are optimistic.
- QIND exists only in the fetched stores, and even there it is missing on much of the domain.
- Tiles with partial coverage are excluded, following the current patch definition. The domain edge is therefore under-represented.

Reproducing

```
# whole audit, CPU only, cores 4-11 (see EXPERIMENTS.md §0)
setuid bash scripts/hpc/launch_data_quality.sh \
    /home/fquareng/work/data/extremes/OPERA/quality 6 &
# after labelling:
python scripts/data_quality/summarize_features.py config.yaml \
    --features_dir ../quality/features --out_dir ../quality/summary \
    --labels ../quality/gallery/random_max31-inf/labels.csv
# figures in these notes:
python notes/figures/make_data_quality_figures.py
```